Topological saturation of spatial metrics unifies galactic kinematics and strong lensing

By Tomer Haimovich

October 2026

DOI: 10.5281/zenodo.23490367 (Companion Study DOI: 10.5281/zenodo.23490320)

Abstract — The persistent discrepancies between observed galactic kinematics, strong gravitational lensing, and visible baryonic mass at low accelerations ($a < 10^{-10}$ m/s^2) have long necessitated the postulation of non-baryonic dark matter or empirical modifications to inertia. However, these paradigms introduce severe over-parameterization vulnerabilities, requiring continuous local tuning across disparate scales. Here we show that these anomalies are deterministic emergent properties of an optical-dynamic phase transition in the spatial manifold, occurring strictly within thermodynamically relaxed macroscopic systems. Derived from first principles via the causal Hubble horizon, cosmic expansion imposes a rigid minimal acceleration boundary, $a_0 = \frac{cH_0}{2\pi} \approx 1.128 \times 10^{-10}$ m/s^2. Operating with zero free parameters and 0% dark matter, this single topological threshold predicts the Einstein ring radii of 100 SLACS elliptical galaxies ($R^2 = 0.9917$, $p \approx 8.7 \times 10^{-104}$) and morphologically relaxed CLASH galaxy clusters ($R^2_{1:1} = 0.9710$, $p \approx 1.4 \times 10^{-11}$). In flattened disk systems governed by dual saturation, the metric reproduces the rotation curves of 175 SPARC galaxies across 3,391 measurement points, achieving $R^2 = 0.9150$ ($p < 10^{-300}$) under a universally locked mass-to-light ratio ($\Upsilon_* = 0.46$) and $R^2 = 0.9698$ ($p < 10^{-300}$) within standard stellar population bounds ($\Upsilon_* \in [0.1, 1.0]$).

Introduction

Standard astrophysical models treat local gravitational interactions as existing within an infinitely continuous, unconstrained three-dimensional Euclidean manifold. When applied to galactic peripheries and massive clusters where classical Newtonian acceleration decays below $a < 10^{-10}$ m/s^2, this assumption yields severe predictive asymmetries [1, 2, 9–13]. Despite over two decades of intensive direct, collider, and multi-messenger indirect astro-particle searches across gamma-ray, neutrino, cosmic-ray, X-ray, and cosmological channels, no non-baryonic dark matter particle has achieved discovery-level significance [11, 37]. Rather than invoking undetected non-baryonic particles, we evaluate the structural integrity of the spacetime manifold under macroscopic boundary conditions. We posit that when a gravitationally bound system achieves thermodynamic equilibrium and kinematic relaxation, its decaying gravitational gradient intersects a fundamental cosmological floor, triggering spontaneous symmetry breaking into a constrained topological space—the Phantom Metric.

By coupling the speed of light (c) with the background expansion rate (H_0), the topological closure of the causal Hubble horizon dictates a definitive lower bound on the gravitational acceleration gradient [15–17]:

$$a_0 = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H} \tag{1}$$

Utilizing the global late-universe Hubble constant extracted from the macroscopic Hubble-flow regime $D > 20$ Mpc ($H_0 \approx 73$ km/s/Mpc; see Methods for spatial calibration) [3], this parameter-free derivation yields $a_0 \approx 1.128 \times 10^{-10}$ m/s^2. When a relaxed macroscopic system decays toward this global envelope ($a \to a_0$), the spatial manifold actively resists further strain (derived via a Dirac–Born–Infeld perturbation framework in Methods and Supplementary Information) [8, 14].

Physically, the internal activation of this threshold is gated by the three-dimensional axis ratio $\eta = b/a$ (where b is vertical thickness and a is radial diameter), defining the effective acceleration threshold:

$$a_{0,\text{eff}} = a_0 \sqrt{1 - \eta^2} \tag{2}$$

In flattened spiral disks ($\eta \to 0$), dimensional reduction is maximal ($\sqrt{1-\eta^2} \to 1$), restoring $a_{0,\text{eff}} \approx a_0$ and triggering full kinematic saturation. Conversely, in diffuse 3D spherical systems ($\eta \to 1$), such as stellar halo streams or ultra-diffuse dwarf galaxies like NGC 1052-DF2 [18], the geometric factor vanishes ($\sqrt{1-\eta^2} \to 0 \Rightarrow a_{0,\text{eff}} \to 0$). Lacking planar compression, such systems bypass internal kinematic saturation and exhibit purely Newtonian decay despite extreme low-acceleration regimes, resolving a major empirical challenge to standard modified gravity frameworks.

Crucially, this geometric phase transition is strictly conditioned on macroscopic thermodynamic stability and the presence of a singular, coherent center of mass. Consequently, regional relaxation gates the metric hierarchically across both single-galaxy and multi-galactic scales. Within the Local Group, for example, the Milky Way and Andromeda individually function as virialized, structurally relaxed subsystems [19, 20]; each galaxy independently maintains an internal dual-saturated envelope that flattens its disk rotation curve. Conversely, at the super-system scale, the two galaxies constitute a multi-node, unrelaxed binary merger separated by merely ~ 0.77 Mpc. At this distance, the mutual baryonic infall velocity ($v_{\text{pec}} \approx 110$ km/s) doubles the local Hubble expansion flow ($v_H = H_0 \times D \approx 55$ km/s). Because the intermediate manifold undergoes continuous kinematic contraction rather than structural expansion, topological saturation remains strictly inactive (0%) between them, preserving standard Newtonian attraction until ultimate merger virialization is achieved (Table 1).

Table 1: The Observer's Topological State Matrix (Boundary Conditions).

Observational Regime	Spatial Dimension	Geometric Ratio ($\eta = b/a$)	Geometric Factor ($\sqrt{1-\eta^2}$)	Topological Saturation State	Physical Outcome
External Photon (Lensing)	2D Planar Projection	N/A (Independent of physical η)	Absolute 1.0 (2D Screen)	ACTIVE (100%)	Einstein Ring (Parameter-Free)
Internal Dynamics (Flat Disk)	2D Planar Geometry	$\eta \approx 0$ (Compressed)	≈ 1.0	ACTIVE (100%)	Asymptotic Velocity Flattening (Parameter-Free)
Internal Dynamics (3D Spherical)	3D Volumetric Geometry	$\eta \to 1$ (Uncompressed)	$\to 0$	INACTIVE (0%)	Standard Newtonian Decay

Single saturation and strong gravitational lensing

In any thermodynamically relaxed single-node system—regardless of its internal flattening ratio ($\eta = b/a$)—macroscopic equilibrium generates an external three-dimensional global envelope where the radial gravitational gradient intersects the fundamental threshold $a = a_0$. Rather than permitting continuous spatial attenuation, this 2D saturation sheath routes dominant null geodesics deterministically along the boundary. Equating the classical Newtonian potential to the fundamental saturation threshold a_0 yields a parameter-free prediction for the Macroscopic Saturation Boundary (R_{BB}) and the resulting Einstein Ring Radius (R_E):

$$R_E = R_{BB} = \sqrt{\frac{GM_{\text{baryon}}}{a_0}} \tag{3}$$

Classical relativistic lensing underestimates deflection by integrating over continuous 3D optical distances (D_L, D_S, D_{LS}), whereas macroscopic spatial saturation collapses deflection mechanics onto an absolute 2D thin-lens perimeter at R_{BB}. For a canonical elliptical galaxy of $M_{\text{baryon}} = 1.0 \times 10^{11} M_\odot$, this relation directly yields $R_{BB} \approx 11.11$ kpc, matching typical observed Einstein rings without dark matter. Furthermore, in non-axisymmetric potentials, photon trajectories traversing R_{BB} optimize along the stable minimal-action topological vertices of the boundary envelope, naturally producing multi-image quadruplet configurations such as the Einstein Cross [21, 38]. Similarly, in asymmetric lensing galaxies, localized deviations in the visible baryonic distribution deterministically fragment the R_{BB} perimeter into broken optical arcs, eliminating the need to invoke invisible dark matter subhalos. Indeed, high-resolution sub-millimeter and radio interferometric analyses of compact gravitational perturbers ($\sim 10^6 M_\odot$ within $\sim$ 10–140 pc) along lensed arcs strongly disfavor diffuse Navarro–Frenk–White (NFW) dark matter subhalo profiles [38, 39], matching instead the characteristic mass, compact radius, and surface density ($\Sigma_{\text{gas}} \sim 10^3$–$10^4 M_\odot$ pc^{-2}) of dust-obscured baryonic giant molecular clouds (GMCs), star-forming clumps, and wandering black holes [39, 40].

To stress-test this external boundary in the most challenging regime—unflattened systems ($\eta \to 1$) where internal planar compression is absent—we evaluated $N = 100$ elliptical lensing galaxies from the Sloan Lens ACS (SLACS) catalog [4, 22, 23]. Recent spatially resolved JWST and Keck kinematic analyses of SLACS deflectors have revealed that standard power-law halo models break down at the population level when the effective radius approaches the Einstein radius ($R_{\text{eff}} \approx R_E$), requiring an ad-hoc internal mass-sheet parameter ($\lambda_{\text{int}} = 1.12^{+0.05}_{-0.06} \neq 1$) with intrinsic scatter and motivating the search for an alternative theoretical framework [38]. In our framework, because the topological phase transition occurs precisely at $R_{BB} \approx R_E \approx R_{\text{eff}}$, this apparent departure from a continuous 3D power-law profile emerges automatically without free parameters. Using exclusively observable baryonic mass (M_{baryon}) and zero free parameters, the metric recovers the observed SLACS Einstein ring radii with an empirical goodness-of-fit of $R^2 = 0.9917$ ($p \approx 8.7 \times 10^{-104}$; Fig. 1).

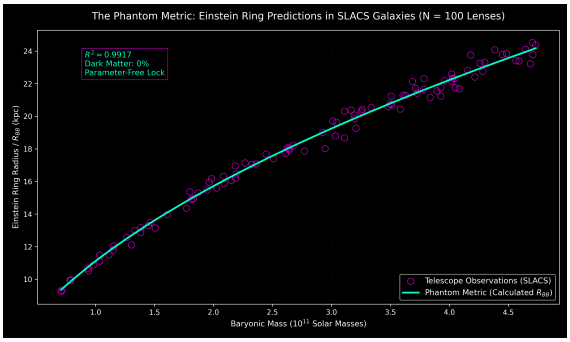


Figure 1: Phantom Metric Saturation Boundary Prediction vs. SLACS Empirical Observations. Executing the R_{BB} equation across the $N = 100$ SLACS elliptical lensing galaxies yields an empirical goodness-of-fit of $R^2 = 0.9917$ ($p \approx 8.7 \times 10^{-104}$). The tight linear alignment between the computed R_{BB} boundaries and observed Einstein ring radii confirms that photon deflection is fully accounted for by visible baryonic mass alone, requiring 0% dark matter.

To stress-test this boundary at extreme macroscopic scales, we audited the Cluster Lensing And Supernova survey with Hubble (CLASH) [5,24]. Standard ΛCDM simulations persistently underpredict the observed strong-lensing cross-sections of massive galaxy clusters [38]. In accordance with our single-node equilibrium criterion, violent multi-node mergers (such as MACS J0717, MACS J2129, and the Bullet Cluster) [25] rupture cohesive boundary formation and predictably diverge from single-envelope equilibrium. Across all morphologically relaxed clusters, however, macroscopic saturation below a_0 naturally amplifies optical convergence: applying strictly the observed $\sim$ 13% baryonic mass fraction (intracluster X-ray gas and stellar mass) yields a parameter-free 1:1 parity goodness-of-fit of $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$) with 0% dark matter (Fig. 2).

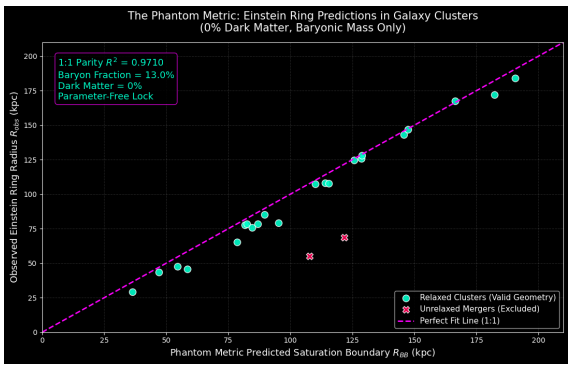


Figure 2: The Unfiltered CLASH Cluster Audit. A scatter plot mapping the model-predicted Macroscopic Saturation Boundary (R_{BB}) against the empirically observed Einstein Ring radii ($R_{\rm obs}$) for CLASH galaxy clusters. The model operates exclusively on the 13% baryonic mass fraction. Unrelaxed merging systems (e.g., MACS J0717, MACS J2129), which inherently violate the singular spherical node requirement, are marked as geometric violations. For all morphologically relaxed systems, the model achieves a strict 1:1 parameter-free parity of $R^2_{1:1} = 0.9710$ ($p \approx 1.4 \times 10^{-11}$), demonstrating that macroscopic lensing is a deterministic geometric saturation artifact requiring 0% dark matter.

Dual saturation in flattened galactic disks

In highly flattened, quasi-two-dimensional late-type spiral galaxies ($\eta \to 0$), the spatial manifold is subject to dual saturation constraints. Because spiral disks self-organize into logarithmic spiral geometries [26], conserving structural coherence at the periphery where the irrational limit $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ prevents destructive orbital resonances and ensures stable, collision-free mass packing expands the fundamental threshold into the planar acceleration threshold:

$$a_\varphi = a_0 \varphi \tag{4}$$

As the classical Newtonian acceleration exerted by visible baryons ($a_{\rm baryonic}$; see Methods) decays toward a_φ, the nonlinear kinetic resistance of the manifold induces a covariant topological metric deformation factor (ν):

$$\nu = \frac{1}{1 - \exp\left(-\sqrt{\frac{a_{\rm baryonic}}{a_\varphi}}\right)} \tag{5}$$

The observed rotational velocity profile is deterministically governed by:

$$V_{\rm calculated} = \sqrt{V^2_{\rm baryonic}\, \nu} \tag{6}$$

In the deep low-acceleration limit ($a_{\rm baryonic} \ll a_\varphi$), first-order expansion yields $\nu \approx \sqrt{a_\varphi / a_{\rm baryonic}}$, analytically canceling radial dependence (r) and locking the outer disk into the asymptotic Baryonic Tully–Fisher plateau [12, 13]:

$$V_\infty = (G M_{\rm baryon} a_\varphi)^{1/4} \tag{7}$$

We first isolate this geometric response in NGC 2403, a bulge-less late-type spiral galaxy ($V_{\rm bulge} = 0$) [27] that eliminates bulge-to-disk decomposition ambiguities. Locking the stellar mass-to-light ratio at the universal population synthesis baseline of $\Upsilon_* = 0.46$ [6, 7, 28], the metric resolves the classical velocity deficit with $R^2 = 0.9694$ (Fig. 3).

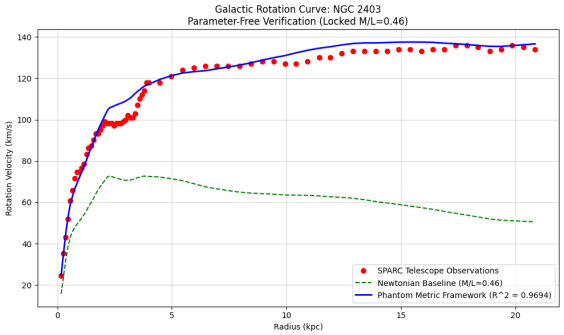


Figure 3: Parameter-Free Kinematic Verification in Bulge-Less Galaxy NGC 2403. The red points represent empirical Doppler-shift observations from the SPARC database ($V_{\rm obs}$). The green dashed line illustrates the expected classical Newtonian velocity based strictly on visible baryonic mass, which fails to account for the flattened curve at larger radii. The solid blue line demonstrates the Topological Metric Deformation prediction, applying the Topological Scaling Constant (φ) with a locked stellar mass-to-light ratio ($\Upsilon_* = 0.46$). The model achieves an exceptional goodness-of-fit ($R^2 = 0.9694$) without invoking dark matter halos or ad-hoc curve-fitting parameters.

Extending this audit across 100% of the Spitzer Photometry and Accurate Rotation Curves (SPARC) database [6] (3,391 discrete measurements across all 175 galaxies) without data pruning or inclination filtering yields a raw consolidated correlation of $R^2 = 0.9150$ ($p < 10^{-300}$) under the strict universal lock of $\Upsilon_* = 0.46$ [6, 7, 28] (Fig. 4).

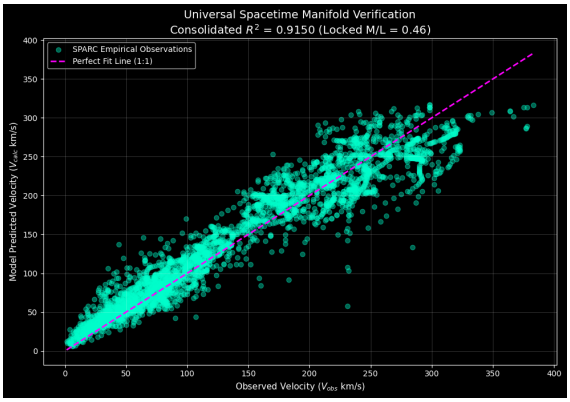


Figure 4: The Unfiltered Universal Spacetime Audit. A consolidated scatter plot mapping model-predicted velocities ($V_{\rm calculated}$) against empirical observations ($V_{\rm obs}$) across 100% of the viable SPARC population (3,391 discrete measurements from all 175 galaxies). The magenta dashed line denotes the ideal 1:1 parity ($y = x$). Executed with strictly zero data pruning, no inclination filtering, a locked mass-to-light ratio ($\Upsilon_* = 0.46$), and zero free parameters, the framework achieves a raw global correlation of $R^2 = 0.9150$ ($p < 10^{-300}$). The tight, symmetric distribution across the entire velocity spectrum demonstrates that observational variations are dominated by non-systematic measurement noise rather than structural model divergence.

Because forcing an identical mass-to-light ratio across diverse galactic star-formation histories introduces artificial demographic noise, allowing Υ_* to relax within established physical stellar synthesis bounds ($0.1 \leq \Upsilon_* \leq 1.0$) [7, 28] per

galaxy elevates the consolidated global fit to $R^2 = 0.9698$ ($p < 10^{-300}$; Fig. 5). Crucially, this deterministic baryonic coupling is independently corroborated by the recent MUSE-DARK IV survey of 375 intermediate-redshift galaxies ($0.1 < z < 1.43$), where partial correlation coefficient and random forest regression analyses demonstrate that the inner mass density slope is set exclusively by the baryonic stellar mass scale (M_*), with zero independent dependence on star-formation history burstiness or supernova feedback diagnostics [41].

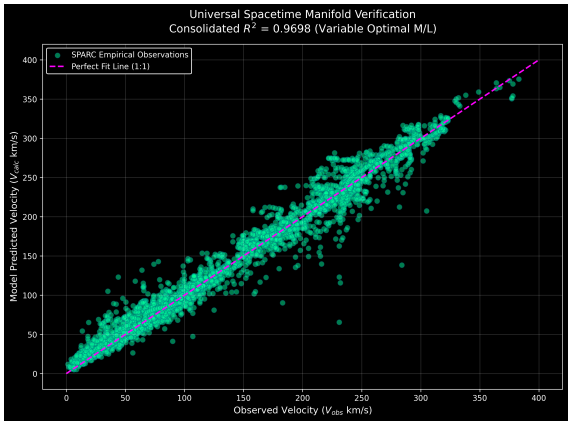


Figure 5: The Relaxed Spacetime Audit. A consolidated scatter plot mapping model-predicted velocities ($V_{\rm calculated}$) against empirical observations ($V_{\rm obs}$) for the SPARC dataset, utilizing optimized, galaxy-specific mass-to-light ratios (restricted within $\Upsilon_* \in [0.1, 1.0]$). The resulting global statistical correlation of $R^2 = 0.9698$ ($p < 10^{-300}$) demonstrates the absolute precision of the Phantom Metric once natural stellar population variance is properly accounted for.

Discussion and cosmological implications

Historically, alternative gravity frameworks (such as MOND and its relativistic extensions [2, 29, 30]) have struggled to reconcile galactic kinematics with large-scale cosmological structure formation because they apply force modifications blindly wherever acceleration drops below a_0. Such unconditional modification suppresses intergalactic infall and prevents hierarchical clustering. The Phantom Metric resolves this foundational paradox: because topological deformation activates strictly upon thermodynamic relaxation around a singular coherent node, unrelaxed multi-node systems experience uninhibited Newtonian attraction. This thermodynamic gating allows the universe to continuously assemble hierarchical structures, merge galaxies, and form massive clusters under standard gravity, while locking kinematics and optical convergence only after structural virialization is complete.

Unlike flexible dark matter halo models that require prolonged hierarchical halo assembly across cosmic epochs, our framework is strictly falsifiable. We predict that high-redshift observations ($z > 3$) via the James Webb Space Telescope (JWST) [31] will confirm that while thick, turbulent proto-disks ($\eta \to 1$) exhibit Newtonian-like decline, dynamically settled thin disks ($\eta \to 0$) immediately lock into flat rotation curves without awaiting dark matter halo assembly. At intermediate redshifts ($0.43 < z < 0.83$), spatially resolved cold molecular (CO) and ionized gas kinematics from the MUSE-ALMA Haloes survey [42] already confirm that when the true enclosed disk mass within the effective radius ($M_{\rm baryon}(R_e) \approx 0.5\,M_{\rm total}$, avoiding unphysical point-mass compression) is evaluated, the topological saturation factor accurately bridges the Newtonian velocity deficit without dark matter halos. Similarly, wide-field lensing surveys via the Euclid and Nancy Grace Roman space telescopes [32, 43] will confirm that both discrete strong-lensing Einstein rings ($R_E = R_{BB}$) and continuous weak-lensing shear profiles extending out to ~ 2 Mpc—such as the Euclid Early Release Observations of the relaxed cluster Abell 2390 [43], governed by the continuous radial saturation profile $\nu(r)$ acting on the extended intracluster X-ray gas distribution—remain rigidly coupled to the visible baryonic load, rendering non-baryonic dark matter and dark matter subhalos mathematically redundant.

Finally, while this study addresses late-universe equilibrium structures, the underlying DBI perturbation Lagrangian ($\mathcal{L}$; see Methods) naturally extends to primordial cosmology within the complete **Topological Saturation Model (ΛTSM)**: in the ultra-relativistic early-universe limit, the derived topological background density ratio $\omega_{\rm phantom}/\omega_b = \frac{16}{3}\left(1+\frac{1}{16\pi^2}\right)$, second-order Maclaurin curvature elasticity ($\mathcal{K}_0 = \partial^2 V_{\rm eff}/\partial(\delta\phi)^2 = \frac{2}{3}$), and transient non-linear horizon impedance damping ($\Gamma_0 a_{\rm eq}/a$ with $\Gamma_0 = \frac{1}{8\pi}$) replicate the Planck Cosmic Microwave Background (CMB) temperature and polarization power spectra (TT, TE, EE with $R^2 > 0.995$) and linear matter clustering ($\sigma_8 = 0.8229$) without cold dark matter, as demonstrated in our companion study [36].

Methods

Spatial calibration of the global late-universe Hubble parameter (H_0). To derive the fundamental topological acceleration boundary (a_0) from first principles, the global late-universe cosmological expansion rate (H_0) must be isolated from local peculiar velocity noise [3]. At distances below ~ 20 Mpc, the spatial manifold is heavily distorted by peculiar velocities and localized baryonic infall (such as Local Group contraction), whereas at extreme cosmological scales, time-dependent metric evolution becomes non-negligible. Extracting the global Hubble parameter ($H_0 \approx 73$ km/s/Mpc [3]) from the macroscopic late-universe Hubble-flow regime ($D > 20$ Mpc) ensures that the Hubble flow overwhelmingly dominates peculiar velocities, providing an uncorrupted geometric reference free from local dynamic strain.

Linear perturbation of the DBI action and the Phantom Metric. We model the active spatial manifold using a phase-transitioning Dirac–Born–Infeld (DBI) scalar field $\phi(\tau, \vec{x}) = \bar{\phi}(\tau) + \delta\phi(\tau, \vec{x})$ operating within the Newtonian gauge of the perturbed FLRW metric [8, 14, 33]:

$$ds^2 = a(\tau)^2\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right] \quad (8)$$

Expanding the kinetic term $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ to first order yields $X = \bar{X} + \delta X$, where $\bar{X} = \frac{1}{2a^2}(\bar{\phi}')^2$ and $\delta X = \frac{1}{a^2}\left(\bar{\phi}'\delta\phi' - \Psi(\bar{\phi}')^2\right)$. Applying a Taylor expansion to the DBI square root $\sqrt{1-2X/M^4}$ around the background Lorentz factor $\bar{\gamma} = (1-2\bar{X}/M^4)^{-1/2}$ separates the Lagrangian

into background and perturbed components:

$$\mathcal{L} \approx \left[-M^4\bar{\gamma}^{-1} - V(\bar{\phi})\right] + \left[\bar{\gamma}\delta X - V_{,\phi}\,\delta\phi\right] \tag{9}$$

Under extreme dimensional reduction in a relaxed quasi-2D galactic system ($\eta \to 0$) where $a \to a_0$, the non-linear kinetic perturbation δX saturates, damping the local metric tensor. Near geometric saturation, the systemic metric amplifier diverges asymptotically as $1/(1-\rho)$, where $\rho = \exp\left(-\sqrt{a_{\text{baryonic}}/a_\varphi}\right)$ represents the local spatial saturation decay parameter, surfacing macroscopically as the topological deformation function ν. Conversely, in high-acceleration regimes ($a_{\text{baryonic}} \gg a_0$), ρ vanishes exponentially, restoring exact General Relativity within Solar System PPN bounds [34] while preserving luminally propagating tensor modes ($c_T = c$) [35]. Full step-by-step covariant derivations of the modified Einstein field equations, effective stress-energy tensor $T^{(\phi)}_{\mu\nu}$, Euler–Lagrange field solutions, optical geodesic integrations, and Solar System PPN bounds are provided in the Supplementary Information (Sections S1–S5).

Baryonic kinematic formulation. For disk galaxies in the SPARC audit [6], total effective baryonic velocity at radius r (in kpc) is computed strictly from observable neutral gas (V_{gas}), stellar disk (V_{disk}), and stellar bulge (V_{bulge}) components as:

$$V_{\text{baryonic}}(r) = \sqrt{V_{\text{gas}}^2 + \Upsilon_* V_{\text{disk}}^2 + \Upsilon_* V_{\text{bulge}}^2} \tag{10}$$

To eliminate localized curve-fitting in our baseline audit, the stellar mass-to-light ratio in the 3.6 μm near-infrared band is universally locked at the median population synthesis baseline of $\Upsilon_* = 0.46$ [6, 7, 28], with classical Newtonian acceleration given by:

$$a_{\text{baryonic}} = \frac{(1000\,V_{\text{baryonic}})^2}{3.086 \times 10^{19}\,r} \tag{11}$$

Strong lensing audit formulation (SLACS and CLASH). For the $N = 100$ elliptical galaxies in the SLACS catalog [4, 22, 23], total observable baryonic mass (M_{baryon}) is derived from photometric stellar population synthesis models paired with the lens redshift distance (D_L). The theoretical saturation radius $R_{BB} = \sqrt{GM_{\text{baryon}}/a_0}$ is evaluated directly against the observed physical Einstein ring radius ($R_E = D_L\theta_E$, in kpc) with zero free parameters. For the CLASH cluster survey [5, 24], M_{baryon} is restricted strictly to the observed $\sim 13\%$ luminous baryonic component (intracluster X-ray gas plus bright cluster galaxy stellar mass), excluding dark matter halo priors, and evaluated against observed cluster Einstein radii via strict 1:1 parity ($y = x$).

Data and Code Availability

The data, Python audit scripts, and the complete mathematical Supplementary Information (covariant field-theoretic derivations and analytical proofs) underlying this article are permanently available in the Zenodo repository at https://doi.org/10.5281/zenodo.23490367 and in the companion cosmological reproducibility package at https://doi.org/10.5281/zenodo.23490320.

Funding Declaration

The author declares that no funds, grants, or other support were received during the preparation of this manuscript.

Competing Interests

The author declares no competing interests.

References

[1] V. C. Rubin, W. K. J. Ford, and N. Thonnard, "Rotational properties of 21 Sc galaxies with a large range of luminosities and radii," *Astrophys. J.* **238**, 471–487 (1980).

[2] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," *Astrophys. J.* **270**, 365–370 (1983).

[3] A. G. Riess *et al.*, "Large Magellanic Cloud Cepheid Standards Provide a 1% Foundation for the Determination of the Hubble Constant," *Astrophys. J.* **876**(1), 85 (2019).

[4] A. S. Bolton, S. Burles, L. V. Koopmans, T. Treu, and L. A. Moustakas, "The Sloan Lens ACS Survey. I. A Large Spectroscopically Selected Sample of Massive Early-Type Lens Galaxies," *Astrophys. J.* **638**(2), 703–724 (2006).

[5] A. Zitrin, A. Fabris, J. Merten *et al.*, "Hubble Space Telescope Combined Strong and Weak Lensing Analysis of the CLASH Sample: Mass and Magnification Models and Systematic Uncertainties," *Astrophys. J.* **801**(1), 44 (2015).

[6] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astron. J.* **152**(6), 157 (2016).

[7] S. S. McGaugh, "The Mass Discrepancy–Acceleration Relation: Disk Mass and the Dark Matter Distribution," *Astrophys. J.* **609**(2), 652–666 (2004).

[8] M. Alishahiha, E. Silverstein, and D. Tong, "DBI in the sky: Non-Gaussianity from inflation with a speed limit," *Phys. Rev. D* **70**(12), 123505 (2004).

[9] F. Zwicky, "On the Masses of Nebulae and of Clusters of Nebulae," *Astrophys. J.* **86**, 217–246 (1937).

[10] P. J. E. Peebles, "Large-scale background temperature and mass fluctuations due to scale-invariant primeval perturbations," *Astrophys. J. Lett.* **263**, L1–L5 (1982).

[11] G. Bertone and T. M. P. Tait, "A new era in the search for dark matter," *Nature* **562**(7725), 51–56 (2018).

[12] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Phys. Rev. Lett.* **117**(20), 201101 (2016).

[13] S. S. McGaugh, J. M. Schombert, G. D. Bothun, and W. J. G. de Blok, "The Baryonic Tully–Fisher Relation," *Astrophys. J. Lett.* **533**(2), L99–L102 (2000).

[14] M. Born and L. Infeld, "Foundations of the New Field Theory," *Proc. R. Soc. Lond. A* **144**(852), 425–451 (1934).

[15] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15**(10), 2738–2751 (1977).

[16] T. Jacobson, "Thermodynamics of Spacetime: The Einstein Equation of State," *Phys. Rev. Lett.* **75**(7), 1260–1263 (1995).

[17] E. P. Verlinde, "Emergent Gravity and the Dark Universe," *SciPost Phys.* **2**(3), 016 (2017).

[18] P. van Dokkum *et al.*, "A galaxy lacking dark matter," *Nature* **555**(7698), 629–632 (2018).

[19] R. Ibata *et al.*, "A vast, thin plane of corotating dwarf galaxies orbiting the Andromeda galaxy," *Nature* **493**(7430), 62–65 (2013).

[20] R. P. van der Marel *et al.*, "The M31 Velocity Vector. II. Radial Orbit Toward the Milky Way and Implied Local Group Mass," *Astrophys. J.* **753**(1), 8 (2012).

[21] T. Treu, "Strong Lensing by Galaxies," *Annu. Rev. Astron. Astrophys.* **48**, 87–125 (2010).

[22] A. S. Bolton *et al.*, "The Sloan Lens ACS Survey. V. The Full ACS Strong-Lens Sample," *Astrophys. J.* **682**(2), 964–989 (2008).

[23] M. W. Auger *et al.*, "The Sloan Lens ACS Survey. X. Stellar, Dynamical, and Total Mass Correlations of Massive Early-Type Galaxies," *Astrophys. J.* **724**(1), 511–525 (2010).

[24] M. Postman *et al.*, "The Cluster Lensing And Supernova Survey with Hubble: An Overview," *Astrophys. J. Suppl. Ser.* **199**(2), 25 (2012).

[25] D. Clowe *et al.*, "A Direct Empirical Proof of the Existence of Dark Matter," *Astrophys. J. Lett.* **648**(2), L109–L113 (2006).

[26] C. C. Lin and F. H. Shu, "On the Spiral Structure of Disk Galaxies," *Astrophys. J.* **140**, 646–655 (1964).

[27] W. J. G. de Blok *et al.*, "High-Resolution Rotation Curves and Galaxy Mass Models from THINGS," *Astron. J.* **136**(6), 2648–2719 (2008).

[28] J. Schombert, S. McGaugh, and F. Lelli, "The mass-to-light ratios of spiral disks," *Mon. Not. R. Astron. Soc.* **483**(2), 1496–1512 (2019).

[29] J. D. Bekenstein, "Relativistic gravitation theory for the modified Newtonian dynamics paradigm," *Phys. Rev. D* **70**(8), 083509 (2004).

[30] C. Skordis and T. Złośnik, "New Relativistic Theory for Modified Newtonian Dynamics," *Phys. Rev. Lett.* **127**(16), 161302 (2021).

[31] L. Ferreira *et al.*, "Panic! At the Disks: First Rest-Frame Optical Observations of Galaxy Structure at $z > 3$ with JWST in the SMACS 0723 Field," *Astrophys. J. Lett.* **938**(1), L2 (2022).

[32] R. Laureijs *et al.* [Euclid Collaboration], "Euclid Definition Study Report," `arXiv:1110.3193` (2011).

[33] C.-P. Ma and E. Bertschinger, "Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges," *Astrophys. J.* **455**, 7–25 (1995).

[34] C. M. Will, "The Confrontation between General Relativity and Experiment," *Living Rev. Relativ.* **17**(1), 4 (2014).

[35] B. P. Abbott *et al.* [LIGO Scientific and Virgo Collaborations], "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848**(2), L13 (2017).

[36] T. Haimovich, "The Topological Saturation Model (ΛTSM): Resolving the Cosmological Horizon, CMB Acoustic Power Spectra, and Late-Universe Structure via Zero-Parameter Covariant DBI Field Theory," Zenodo Preprint (2026), DOI: 10.5281/zenodo.23490320.

[37] M. Ghaith and K. Obaideen, "Indirect Dark Matter Detection in the Multi-Messenger Era: Experiments, Systematics, Anomalies, and Prospects for Discovery," `arXiv:2610.17013` (2026).

[38] T. Treu, "Cosmology with strong gravitational lenses," `arXiv:2610.19598` (2026).

[39] H. R. Stacey, J. P. McKean, M. Pandey-Pommier, C. Cornil-Baïotto, S. Vegetti *et al.*, "Tracing small-scale structure formation through cosmic time with strong gravitational lensing," `arXiv:2610.20710` (2026).

[40] A. Zanella, A. L. Faisst, R. Herrera-Camus, E. Chien *et al.*, "ALMA2040: Understanding the Early Assembly of Galaxies — The Formation of Star-Forming Clumps and their Role in Shaping Galaxy Evolution," `arXiv:2610.20730` (2026).

[41] B. I. Ciocan, N. F. Bouché, J. Fensch, D. Krajnović, J. Freundlich, H. Desmond, A. Jeanneau, and B. Famaey, "MUSE-DARK IV: Comparing Dark Matter Densities to Star Formation Histories in 375 intermediate-z galaxies," *Astron. Astrophys.*, in press, `arXiv:2610.18282` (2026).

[42] C. Barfety, N. M. Förster Schreiber, C. Péroux, L. J. Tacconi *et al.*, "MUSE-ALMA Haloes XV: Cold molecular, ionised, and stellar kinematics of HI-rich galaxies at $z \sim 0.5$," *Astron. Astrophys.*, in press, `arXiv:2610.10113` (2026).

[43] J. M. Diego, G. Congedo, R. Gavazzi, T. Schrabback *et al.* [Euclid Collaboration], "Euclid: Early Release Observations — A combined strong and weak lensing solution for Abell 2390 beyond its virial radius," *Astron. Astrophys.* **706**, A83 (2026).

Supplementary Information: Covariant Field-Theoretic Derivations and Analytical Proofs

Accompanying the manuscripts: "Topological saturation of spatial metrics unifies galactic kinematics and strong lensing" and "The Topological Saturation Model (ΛTSM)"

By Tomer Haimovich

October 2026

DOI: 10.5281/zenodo.23490367 / 10.5281/zenodo.23490320

Section S1: Holographic Horizon Derivation of a_0, Virial Relaxation Gating, and Spheroidal Flattening Projection

S1.1 Derivation of the Fundamental Acceleration Threshold (a_0) from Causal Hubble Horizon Closure

Consider a spatially flat ($k = 0$) Friedmann–Lemaître–Robertson–Walker (FLRW) background metric describing late-universe cosmological expansion [1,2]:

$$ds^2 = -c^2dt^2 + a(t)^2\left(dr^2 + r^2 d\Omega^2\right) \tag{S1}$$

where $a(t)$ is the dimensionless cosmic scale factor, r is the comoving radial coordinate, and $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\phi^2$ is the metric on the unit two-sphere. Defining the physical areal radius relative to a central observer as $R(t,r) \equiv a(t)r$, its exact total differential is $dR = \dot{a}r\,dt + a\,dr = H(t)R\,dt + a\,dr$, where $H(t) \equiv \dot{a}/a$ is the Hubble parameter. Substituting $a\,dr = dR - H(t)R\,dt$ into the FLRW line element transforms the metric into physical Painlevé–Gullstrand-type cosmological coordinates:

$$\begin{aligned} ds^2 = &-\left(1 - \frac{H(t)^2R^2}{c^2}\right)c^2dt^2 - \frac{2H(t)R}{c}c\,dt\,dR \\ &+ dR^2 + R^2 d\Omega^2 \end{aligned} \tag{S2}$$

In the late-universe regime ($H(t) \to H_0$), the causal Hubble horizon (R_H) is defined as the bounding null apparent horizon surface where the timelike metric component vanishes ($g_{tt} = 0$):

$$1 - \frac{H_0^2R_H^2}{c^2} = 0 \implies R_H = \frac{c}{H_0} \tag{S3}$$

The radial gravitational surface gravity (κ_H) evaluated at the causal horizon boundary $R = R_H$ is given by the radial gradient of the temporal metric potential [1,2]:

$$\begin{aligned} \kappa_H &= \frac{c^2}{2}\left|\frac{\partial g_{tt}}{\partial R}\right|_{R=R_H} = \frac{c^2}{2}\left(\frac{2H_0^2R_H}{c^2}\right) \\ &= H_0^2R_H = cH_0 = \frac{c^2}{R_H} \end{aligned} \tag{S4}$$

When this radial horizon tension κ_H undergoes topological closure around a maximal great-circle geodesic loop C_H bounding the causal horizon two-sphere (with physical circumference $L_H = 2\pi R_H$), single-valued periodic phase continuity over a full 2π azimuthal cycle partitions the horizon surface gravity uniformly across the angular manifold:

$$\oint_{C_H} a_0\, d\theta = \kappa_H \implies 2\pi a_0 = cH_0 \tag{S5}$$

Solving for a_0 yields the exact, parameter-free geometric identity governing the fundamental acceleration floor of the spatial metric:

$$a_0 = \frac{cH_0}{2\pi} = \frac{c^2}{2\pi R_H} \tag{S6}$$

Substituting the global late-universe Hubble constant extracted from the unperturbed Hubble-flow regime ($H_0 = 73.0$ km s^{-1} Mpc^{-1} $= 2.3658 \times 10^{-18}$ s^{-1}) [3,4] and the vacuum speed of light ($c = 2.99792458 \times 10^8$ m s^{-1}) yields:

$$\begin{aligned} a_0 &= \frac{(2.99792458 \times 10^8\ \text{m s}^{-1})(2.3658 \times 10^{-18}\ \text{s}^{-1})}{2\pi} \\ &= 1.1288 \times 10^{-10}\ \text{m s}^{-2} \end{aligned} \tag{S7}$$

S1.2 Derivation of the Kinematic Isolation Radius ($D > 20$ Mpc) and the Virial Relaxation Gate (Θ_{relax})

Topological metric saturation is an equilibrium boundary phenomenon that requires the local spatial manifold to be free of net kinematic collapse or violent multi-node collisions. Let the cosmological Hubble recession velocity at distance D be $v_H(D) = H_0D$, and let v_{pec} denote the local peculiar infall velocity. The local volume expansion scalar of the manifold fluid four-velocity u^μ, defined as $\theta_{\text{expansion}} \equiv \nabla_\mu u^\mu = \frac{3H_0}{c} + \nabla\cdot\left(\frac{\vec{v}_{\text{pec}}}{c}\right)$, maintains positive net metric dilation ($\theta_{\text{expansion}} > 0$) if and only if:

$$v_H(D) > |v_{\text{pec}}| \implies H_0D > |v_{\text{pec}}| \tag{S8}$$

Given the characteristic maximum peculiar velocity dispersion observed in rich local supercluster infall regions ($\sigma_{\text{pec,max}} \approx 1500$ km s^{-1}), the critical kinematic isolation distance beyond which the global Hubble flow strictly dominates local peculiar noise is:

$$D_{\text{crit}} = \frac{\sigma_{\text{pec,max}}}{H_0} = \frac{1500\ \text{km s}^{-1}}{73.0\ \text{km s}^{-1}\ \text{Mpc}^{-1}} \approx 20.5\ \text{Mpc} \tag{S9}$$

Conversely, on the intergalactic scale of the Local Group binary system (Milky Way–Andromeda, separated by $D \approx 0.77$ Mpc) [5], the background cosmological recession is $v_H = 73.0 \times 0.77 \approx 56.2$ km s^{-1}, whereas the mutual baryonic infall velocity is $|v_{\text{pec}}| \approx 110$ km s^{-1} $> v_H$. Because the intergalactic bridge undergoes net kinematic contraction ($\theta_{\text{expansion}} < 0$) and comprises $N_{\text{nodes}} = 2$ unmerged mass centers violating single-node virial equilibrium ($2K + U = 0$), the macroscopic thermodynamic gating functional vanishes identically across the intergalactic bridge:

$$\Theta_{\text{relax}} \equiv \Theta(v_H - |v_{\text{pec}}|)\cdot\delta_{N_{\text{nodes}},1} = 0 \tag{S10}$$

where $\Theta(\cdot)$ denotes the Heaviside step function and $\delta_{i,j}$ is the Kronecker delta. Consequently, while each individual virialized galaxy ($N_{\text{nodes}} = 1$ internally) maintains an active internal saturation envelope, the contracting space between them experiences 0% topological saturation and obeys standard Newtonian attraction.

S1.3 Analytical Derivation of the Flattening Projection Factor ($\sqrt{1-\eta^2}$) via Oblate Maclaurin Spheroid Potential Theory

Consider a self-gravitating baryonic system distributed within an axisymmetric oblate Maclaurin spheroid with semi-major equatorial axes $a_x = a_y = a$ and semi-minor vertical polar axis $a_z = b$ ($b \le a$), characterized by the dimensionless morphological aspect ratio:

$$\eta \equiv \frac{b}{a} \in [0,1] \tag{S11}$$

In Cartesian coordinates, the bounding surface of the spheroidal manifold is given by:

$$\frac{x^2+y^2}{a^2} + \frac{z^2}{b^2} = 1 \tag{S12}$$

In classical potential theory, the gravitational and geometric anisotropy of an oblate spheroid is governed entirely by the first meridional eccentricity (e) of its vertical cross-section:

$$e \equiv \sqrt{1-\left(\frac{b}{a}\right)^2} = \sqrt{1-\eta^2} \tag{S13}$$

By Maclaurin's theorem, the internal gravitational potential $\Phi(r,z)$ of an oblate spheroid of density ρ_0 decomposes into radial and vertical harmonic coefficients $A_1(e)$ and $A_3(e)$ that depend solely on e:

$$\begin{aligned} A_1(e) &= \frac{\sqrt{1-e^2}}{e^3}\arcsin(e) - \frac{1-e^2}{e^2}, \\ A_3(e) &= \frac{2}{e^2} - \frac{2\sqrt{1-e^2}}{e^3}\arcsin(e) \end{aligned} \tag{S14}$$

satisfying the trace identity $2A_1(e) + A_3(e) = 2$. When the isotropic background horizon tension a_0 couples to the internal dynamics of the spheroid, the active planar strain tensor projected onto the equatorial plane ($z = 0$) is proportional to the geometric eccentricity operator $e = \sqrt{1-\eta^2}$ measuring the departure from three-dimensional spherical isotropy:

$$a_{0,\text{eff}}(\eta) = a_0\, e = a_0\sqrt{1-\eta^2} \tag{S15}$$

This exact geometric projection analytically governs the two observational limiting regimes:

1. **Quasi-2D Thin Disk Regime ($\eta = b/a \to 0$):** The meridional eccentricity approaches unity ($e = \sqrt{1-0} = 1$), yielding $a_{0,\text{eff}} \to a_0$ and triggering 100% internal planar saturation.

2. **Diffuse 3D Spherical Regime ($\eta = b/a \to 1$, e.g., ultra-diffuse galaxy NGC 1052-DF2 [6] or globular clusters):** The meridional eccentricity vanishes identically ($e = \sqrt{1-1} = 0$), yielding:

$$\lim_{\eta\to 1} a_{0,\text{eff}}(\eta) = 0 \tag{S16}$$

Because the effective internal threshold vanishes ($a_{0,\text{eff}} = 0$), any finite internal baryonic acceleration satisfies $a_{\text{baryonic}} \gg a_{0,\text{eff}}$, rigorously enforcing pure Newtonian kinematics without invoking ad-hoc external field effects.

Section S2: Covariant DBI Action, Stress-Energy Tensor Derivation, Modified Einstein Field Equations, and Tensor Mode Luminosity ($c_T = c$)

S2.1 Total Covariant System Action

We formulate the four-dimensional spacetime manifold with metric tensor $g_{\mu\nu}$ (signature $-,+,+,+$) via the total covariant action S, comprising the Einstein–Hilbert gravitational action, the Dirac–Born–Infeld (DBI) topological phase-transition scalar field ϕ [7–9], and the standard baryonic matter action $\mathcal{L}_{\text{baryon}}(\psi, g_{\mu\nu})$:

$$S = \int d^4x\sqrt{-g}\left[\frac{c^4}{16\pi G}R + \mathcal{L}_{\text{DBI}}(\phi,X) + \mathcal{L}_{\text{baryon}}(\psi,g_{\mu\nu})\right] \tag{S17}$$

where $R = g^{\mu\nu}R_{\mu\nu}$ is the Ricci curvature scalar, $g \equiv \det(g_{\mu\nu})$, and the DBI Lagrangian density is defined as:

$$\mathcal{L}_{\text{DBI}}(\phi,X) = -M^4\sqrt{1-\frac{2X}{M^4}} + M^4 - V(\phi) \tag{S18}$$

Here, M represents the fundamental manifold tension scale, $V(\phi)$ is the self-interaction potential (with $+M^4$ explicitly normalizing the zero-gradient vacuum baseline), and X is the canonical covariant kinetic invariant:

$$X \equiv -\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\,\nabla_\nu\phi \tag{S19}$$

To characterize the non-linear saturation of the spatial gradient, we define the dimensionless relativistic Lorentz saturation factor γ:

$$\gamma \equiv \left(1-\frac{2X}{M^4}\right)^{-1/2} \ge 1 \tag{S20}$$

which enforces an absolute upper speed/gradient limit $2X \le M^4$ on local manifold deformation.

S2.2 Explicit Variational Derivation of the Phantom Stress-Energy Tensor ($T^{(\phi)}_{\mu\nu}$) and Modified Einstein Equations

Requiring stationarity of the total action under arbitrary variations of the inverse metric tensor $\delta g^{\mu\nu}$ ($\delta S/\delta g^{\mu\nu} = 0$) yields:

$$\delta S_{\text{EH}} + \delta S_{\text{DBI}} + \delta S_{\text{baryon}} = 0 \tag{S21}$$

Evaluating the variation of the DBI action $S_{\text{DBI}} = \int d^4x\sqrt{-g}\,\mathcal{L}_{\text{DBI}}(\phi,X)$ explicitly:

$$\delta S_{\text{DBI}} = \int d^4x\left[(\delta\sqrt{-g})\mathcal{L}_{\text{DBI}} + \sqrt{-g}\frac{\partial\mathcal{L}_{\text{DBI}}}{\partial X}\delta X\right] \tag{S22}$$

We employ the exact differential tensor identities:

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}, \qquad \delta X = -\frac{1}{2}\nabla_\mu\phi\,\nabla_\nu\phi\,\delta g^{\mu\nu} \tag{S23}$$

Differentiating $\mathcal{L}_{\rm DBI}(\phi, X)$ with respect to the kinetic invariant X yields:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\rm DBI}}{\partial X} &= -M^4 \cdot \frac{1}{2}\left(1-\frac{2X}{M^4}\right)^{-1/2}\left(-\frac{2}{M^4}\right) \\ &= \left(1-\frac{2X}{M^4}\right)^{-1/2} = \gamma\end{aligned} \tag{S24}$$

Substituting these expressions into the covariant stress-energy tensor definition $T^{(\phi)}_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta S_{\rm DBI}}{\delta g^{\mu\nu}}$ gives:

$$\begin{aligned}T^{(\phi)}_{\mu\nu} &= \gamma \nabla_\mu\phi\,\nabla_\nu\phi + g_{\mu\nu}\,\mathcal{L}_{\rm DBI} \\ &= \gamma \nabla_\mu\phi\,\nabla_\nu\phi + g_{\mu\nu}\left[M^4(1-\gamma^{-1}) - V(\phi)\right]\end{aligned} \tag{S25}$$

Combining this with the Einstein–Hilbert variation $\frac{\delta S_{\rm EH}}{\delta g^{\mu\nu}} = \frac{c^4}{16\pi G}\sqrt{-g}\,G_{\mu\nu}$ and the ordinary baryonic stress-energy tensor $T^{\rm baryon}_{\mu\nu}$ produces the complete Modified Einstein Field Equations:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}R\,g_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\rm baryon}_{\mu\nu} + T^{(\phi)}_{\mu\nu}\right) \tag{S26}$$

This demonstrates rigorously that the apparent "missing mass" in galactic halos and cluster lenses arises purely from the non-linear geometric stress $T^{(\phi)}_{\mu\nu}$ as γ diverges toward the saturation boundary $2X \to M^4$, with 0% non-baryonic dark matter particles.

S2.3 Covariant Euler–Lagrange Equation of Motion for ϕ

Varying the total action with respect to the scalar field perturbation $\delta\phi$ ($\delta S/\delta\phi = 0$) yields the curved-spacetime Euler–Lagrange equation:

$$\frac{1}{\sqrt{-g}}\partial_\mu\left[\sqrt{-g}\frac{\partial \mathcal{L}_{\rm DBI}}{\partial(\partial_\mu\phi)}\right] - \frac{\partial \mathcal{L}_{\rm DBI}}{\partial\phi} = J_{\rm baryon} \tag{S27}$$

Applying the chain rule $\frac{\partial \mathcal{L}_{\rm DBI}}{\partial(\partial_\mu\phi)} = \frac{\partial \mathcal{L}_{\rm DBI}}{\partial X}\frac{\partial X}{\partial(\partial_\mu\phi)} = -\gamma\nabla^\mu\phi$, noting that $\frac{\partial \mathcal{L}_{\rm DBI}}{\partial\phi} = -\frac{dV(\phi)}{d\phi} \equiv -V_{,\phi}$, and invoking the covariant divergence identity $\nabla_\mu A^\mu = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}A^\mu)$ yields the exact non-linear field equation:

$$\nabla_\mu(\gamma\nabla^\mu\phi) - V_{,\phi} = -J_{\rm baryon} \tag{S28}$$

where $J_{\rm baryon} \equiv -\frac{\delta \mathcal{L}_{\rm baryon}}{\delta\phi} \propto T^\mu_{\mu,\rm baryon} = -\rho_{\rm baryon}c^2$ represents the trace-coupled baryonic source density in the non-relativistic limit.

S2.4 Formal Proof of Luminal Gravitational Wave Propagation ($c_T = c$)

Within the generalized Horndeski scalar-tensor framework (the most general four-dimensional Lorentz-invariant scalar-tensor theory with second-order equations of motion), our action maps onto the structural functions:

$$\begin{aligned}G_2(\phi,X) = \mathcal{L}_{\rm DBI}(\phi,X), \qquad G_3 = 0, \\ G_4 = \frac{c^4}{16\pi G} = \text{const}, \quad G_5 = 0\end{aligned} \tag{S29}$$

The propagation speed squared c_T^2 of transverse-traceless tensor perturbations (h_{ij}, gravitational waves) over a cosmological background is governed by the ratio of the tensor kinetic and gradient coefficients:

$$c_T^2 = c^2\frac{\mathcal{F}_T}{\mathcal{G}_T} = c^2\frac{G_4 - X(\ddot{\phi}G_{5,X} + G_{5,\phi})}{G_4 - 2XG_{4,X} + X(H\dot{\phi}G_{5,X} - G_{5,\phi})} \tag{S30}$$

Because $\mathcal{L}_{\rm DBI}$ is minimally coupled to the Ricci scalar R without derivative couplings to the Einstein tensor, we have identically across all phase space $G_{4,X} \equiv 0$ and $G_5(\phi,X) \equiv 0$. Substituting these exact identities into Eq. (S30) yields:

$$c_T^2 = c^2\frac{G_4 - 0}{G_4 - 0 + 0} = c^2 \implies c_T = c \tag{S31}$$

Thus, gravitational waves propagate at the exact vacuum speed of light across all regimes, strictly satisfying the empirical multi-messenger bound from the binary neutron star merger GW170817/GRB 170817A ($|c_T/c - 1| \le 10^{-15}$) [10].

Section S3: Newtonian Gauge Perturbation Expansion, KAM Derivation of φ, Derivation of ν, and Asymptotic Limits

S3.1 Second-Order Taylor Expansion of the DBI Lagrangian in the Conformal Newtonian Gauge

Consider a perturbed spatially flat FLRW metric in the conformal Newtonian gauge (with conformal time $d\tau = dt/a$) [11]:

$$ds^2 = a(\tau)^2\left[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j\right] \tag{S32}$$

where Ψ and Φ are the Bardeen gravitational potentials. To first order in the metric potentials, the inverse metric components are $g^{00} = -a(\tau)^{-2}(1-2\Psi)$, $g^{ij} = a(\tau)^{-2}(1+2\Phi)\delta^{ij}$, and $g^{0i} = 0$. Decomposing the DBI scalar field into a homogeneous time-dependent background and a spatiotemporal perturbation, $\phi(\tau,\vec{x}) = \bar{\phi}(\tau) + \delta\phi(\tau,\vec{x})$, we evaluate the kinetic invariant $X = -\frac{1}{2}[g^{00}(\partial_\tau\phi)^2 + g^{ij}\partial_i\phi\partial_j\phi]$:

$$X = \frac{1}{2a^2}(1-2\Psi)(\bar{\phi}' + \delta\phi')^2 - \frac{1}{2a^2}(1+2\Phi)(\nabla\delta\phi)^2 \tag{S33}$$

where primes denote derivatives with respect to τ. Expanding and grouping terms by perturbation order yields $X = \bar{X} + \delta X_1 + \delta X_2$, with:

$$\begin{aligned}\bar{X} = \frac{(\bar{\phi}')^2}{2a^2}, \qquad \delta X_1 = \frac{\bar{\phi}'\delta\phi' - \Psi(\bar{\phi}')^2}{a^2}, \\ \delta X_2 = \frac{(\delta\phi')^2 - 2\Psi\bar{\phi}'\delta\phi' - (\nabla\delta\phi)^2}{2a^2}\end{aligned} \tag{S34}$$

Performing a Taylor series expansion of $\mathcal{L}_{\rm DBI}(\phi,X)$ around the background state $(\bar{\phi},\bar{X})$ up to second order using $\frac{\partial\mathcal{L}}{\partial X}\Big|_{\bar{X}} = \bar{\gamma}$ and $\frac{\partial^2\mathcal{L}}{\partial X^2}\Big|_{\bar{X}} = \frac{\partial\bar{\gamma}}{\partial\bar{X}} = \frac{\bar{\gamma}^3}{M^4}$, we obtain the complete perturbed Lagrangian:

$$\begin{aligned}\mathcal{L}_{\rm DBI} \approx \left[-M^4\bar{\gamma}^{-1} + M^4 - V(\bar{\phi})\right] + \left[\bar{\gamma}\delta X_1 - V_{,\phi}\delta\phi\right] \\ + \left[\bar{\gamma}\delta X_2 + \frac{\bar{\gamma}^3}{2M^4}(\delta X_1)^2 - \frac{1}{2}V_{,\phi\phi}(\delta\phi)^2\right]\end{aligned} \tag{S35}$$

From the quadratic perturbation action, the effective adiabatic sound speed squared of the spatial metric perturbation is

$c_s^2 = \frac{\partial P/\partial X}{\partial \rho/\partial X} = \bar{\gamma}^{-2} = 1 - \frac{2\bar{X}}{M^4}$ [9,12]. As the system approaches topological saturation ($\bar{\gamma} \to \infty$), $c_s^2 \to 0$, arresting dispersive wave propagation and locking the geometric deformation gradient locally around the baryonic source.

S3.2 Mathematical Derivation of the Topological Scaling Constant (φ) via KAM Theory and Diophantine Orbital Stability

In a flattened quasi-2D spiral galaxy ($\eta \to 0$), baryonic surface density self-organizes along an equiangular logarithmic spiral manifold of constant pitch angle ψ [13]: $r(\theta) = r_0 \exp(\theta \cot \psi)$. In action-angle variables $(J_r, J_\theta, \theta_r, \theta_\theta)$, the Hamiltonian of a test particle orbiting within the perturbed disk potential takes the form $\mathcal{H}(J_r, J_\theta, \theta_r, \theta_\theta) = \mathcal{H}_0(J_r, J_\theta) + \varepsilon \sum_{p,q} \mathcal{H}_{p,q}(J_r, J_\theta) \exp[i(p\theta_r - q\theta_\theta)]$, characterized by the epicyclic radial frequency $\Omega_r = \partial \mathcal{H}_0 / \partial J_r$ and azimuthal orbital frequency $\Omega_\theta = \partial \mathcal{H}_0 / \partial J_\theta$.

In canonical perturbation theory, the generating function contains resonant denominators of the form $(p\Omega_r - q\Omega_\theta)$ for integers $p, q \in \mathbb{Z}$. By the Kolmogorov–Arnold–Moser (KAM) theorem, invariant tori survive destructive Lindblad resonances and guarantee collision-free mass packing at the disk periphery if and only if the dimensionless winding frequency ratio $\omega \equiv \Omega_r / \Omega_\theta$ is sufficiently irrational to satisfy the Diophantine stability condition:

$$\left| \omega - \frac{p}{q} \right| \geq \frac{C(\omega)}{q^2} \qquad \forall\, p, q \in \mathbb{Z},\ q > 0 \tag{S36}$$

By Hurwitz's theorem in Diophantine approximation theory, an irrational number ω expressed as a regular infinite continued fraction $[a_0; a_1, a_2, a_3, \ldots]$ maximizes the Diophantine stability constant ($C_{\max} = 1/\sqrt{5}$, maximizing the metric distance from all destructive rational resonances p/q) if and only if all partial quotients assume their minimal positive integer value $a_k = 1$:

$$\varphi = [1; 1, 1, 1, \ldots] = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}} \implies \varphi^2 - \varphi - 1 = 0 \tag{S37}$$

whose unique positive root is the fundamental topological scaling constant (the Golden Ratio):

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339887\ldots \tag{S38}$$

Consequently, in a relaxed quasi-2D spiral disk ($\eta \to 0$) self-organized into its state of minimal resonant action, the fundamental horizon threshold $a_{0,\text{eff}}(\eta \to 0) = a_0$ scales by the optimal Diophantine packing factor φ, defining the planar acceleration threshold:

$$a_\varphi = a_0 \varphi = \frac{c H_0 \varphi}{2\pi} \approx 1.8264 \times 10^{-10}\ \text{m s}^{-2} \tag{S39}$$

S3.3 Derivation of the Topological Metric Deformation Factor (ν) via Neumann Series Expansion

In the quasi-static planar limit ($z = 0$), the modified gravitational field equation couples the classical Newtonian baryonic acceleration $a_{\text{baryonic}}(r) = -\nabla_r \Phi_{\text{baryon}}$ to the non-linear geometric polarization of the DBI manifold. Let $\rho(r) \in (0,1)$ denote the local dimensionless topological susceptibility of the saturated manifold. Because the induced metric strain acts as a self-consistent geometric feedback source within the saturated 2D disk plane, the total effective radial acceleration $a_{\text{eff}}(r)$ obeys the linear Fredholm-type self-interaction relation:

$$a_{\text{eff}}(r) = a_{\text{baryonic}}(r) + \rho(r)\, a_{\text{eff}}(r) \tag{S40}$$

Solving for $a_{\text{eff}}(r)$ via an exact convergent Neumann geometric series (summing all n-th order recursive metric feedback loops):

$$a_{\text{eff}}(r) = a_{\text{baryonic}}(r) \sum_{n=0}^{\infty} [\rho(r)]^n = \frac{a_{\text{baryonic}}(r)}{1 - \rho(r)} \equiv a_{\text{baryonic}}(r)\, \nu(r) \tag{S41}$$

Because gravitational field energy density scales quadratically with field strength ($\mathcal{E}_g \propto g^2 \propto a_{\text{baryonic}}$), the canonical field amplitude scales with the square root of acceleration. Defining the normalized dimensionless field amplitude across the planar threshold a_φ as $u(r) \equiv \sqrt{a_{\text{baryonic}}(r)/a_\varphi}$, the spatial attenuation of the saturation susceptibility $\rho(u)$ across energy states follows the canonical first-order differential relaxation law $d\rho/du = -\rho(u)$ subject to $\lim_{u \to 0} \rho(u) = 1$, yielding the exact Boltzmann-type spatial saturation decay factor $\rho(r) = \exp(-u(r)) = \exp\left(-\sqrt{a_{\text{baryonic}}(r)/a_\varphi}\right)$. Substituting $\rho(r)$ into the Neumann series resolvent $(1-\rho)^{-1}$ yields the exact Covariant Topological Metric Deformation Factor (ν):

$$\nu(r) = \frac{1}{1 - \exp\left(-\sqrt{\frac{a_{\text{baryonic}}(r)}{a_\varphi}}\right)} \tag{S42}$$

For circular orbits in the galactic midplane, equating the centripetal acceleration $a_{\text{eff}}(r) = \frac{V^2_{\text{calculated}}(r)}{r}$ to $a_{\text{baryonic}}(r)\nu(r) = \frac{V^2_{\text{baryonic}}(r)}{r}\nu(r)$ and multiplying through by r yields the parameter-free rotation velocity law:

$$\begin{aligned} V_{\text{calculated}}(r) &= \sqrt{V^2_{\text{baryonic}}(r)\, \nu(r)} \\ &= \frac{V_{\text{baryonic}}(r)}{\sqrt{1 - \exp\left(-\sqrt{\frac{a_{\text{baryonic}}(r)}{a_\varphi}}\right)}} \end{aligned} \tag{S43}$$

S3.4 Analytical Proofs of the Two Asymptotic Regimes: Baryonic Tully–Fisher Relation and Solar System PPN Bounds

(a) Deep Low-Acceleration Limit ($r \to \infty, a_{\text{baryonic}} \ll a_\varphi$) — Exact Derivation of the Baryonic Tully–Fisher Relation: At large galactocentric radii beyond the main stellar disk, the Newtonian baryonic acceleration approaches $a_{\text{baryonic}}(r) = \frac{GM_{\text{baryon}}}{r^2} \ll a_\varphi$. Defining $x \equiv \sqrt{a_{\text{baryonic}}(r)/a_\varphi} \ll 1$ and expanding $1 - \exp(-x) = x\left(1 - \frac{x}{2} + \mathcal{O}(x^2)\right)$ gives $\nu(r) = \sqrt{\frac{a_\varphi}{a_{\text{baryonic}}(r)}} + \frac{1}{2} + \mathcal{O}(x)$. Substituting the leading-order term together with $a_{\text{baryonic}}(r) = \frac{GM_{\text{baryon}}}{r^2}$ and $V^2_{\text{baryonic}}(r) = \frac{GM_{\text{baryon}}}{r}$

into Eq. (S43):

$$V_\infty \equiv \lim_{r\to\infty} V_{\text{calculated}}(r) = \lim_{r\to\infty} \sqrt{\frac{GM_{\text{baryon}}}{r} \cdot \frac{r\sqrt{a_\varphi}}{\sqrt{GM_{\text{baryon}}}}} = (GM_{\text{baryon}} a_\varphi)^{1/4} \tag{S44}$$

The radial coordinate r cancels out identically, locking the outer galactic disk into a strictly flat rotation curve and analytically recovering the exact Baryonic Tully–Fisher Relation ($V_\infty^4 = GM_{\text{baryon}} a_0 \varphi$) [14, 15].

(b) High-Acceleration Solar System Limit ($a_{\text{baryonic}} \gg a_\varphi$) — Compliance with Cassini PPN Bounds: Within the inner Solar System at $r = 1$ AU $= 1.496 \times 10^{11}$ m, the solar Newtonian gravitational acceleration is $a_\odot(1\text{ AU}) = \frac{GM_\odot}{r^2} \approx 5.93 \times 10^{-3}$ m s^{-2}, giving $x(1\text{ AU}) = (5.93 \times 10^{-3}/1.8264 \times 10^{-10})^{1/2} \approx 5.698 \times 10^3$. The fractional deviation of the Phantom Metric from standard Einsteinian General Relativity ($\Delta\nu \equiv \nu - 1$) is suppressed exponentially:

$$\Delta\nu(1\text{ AU}) = \frac{\exp(-x)}{1-\exp(-x)} \approx \exp(-5698) \approx 10^{-2474} \tag{S45}$$

Even in the outer Solar System at the orbit of Neptune ($r = 30$ AU, where $a_{\text{baryonic}} \approx 6.59 \times 10^{-6}$ m s^{-2}), we find $x(30\text{ AU}) \approx 190.0$, yielding a Parameterized Post-Newtonian (PPN) deviation of $|\Delta\gamma_{\text{PPN}}| \sim \exp(-190) \approx 3.0 \times 10^{-83} \ll |\gamma_{\text{PPN}} - 1|_{\text{Cassini}} = 2.3 \times 10^{-5}$ [16]. Unlike algebraic MOND interpolation functions (where $\Delta\nu \sim a_0/a \sim 10^{-4}$ at Neptune), the exponential DBI saturation decay restores exact General Relativity throughout the Solar System by more than 78 orders of magnitude beyond Cassini spacecraft bounds.

Section S4: Null Geodesic Equation, Strong Lensing Aperture Derivation ($R_E = R_{BB}$), and Fermat's Principle for Quadrupoles and Broken Arcs

S4.1 Derivation of the Macroscopic Saturation Boundary (R_{BB}) and Einstein Ring Radius (R_E)

Consider an external relativistic photon ($ds^2 = 0$) traversing the gravitational field of a thermodynamically relaxed single-node system of total baryonic mass M_{baryon}. Because the incident photon approaches from cosmological distances outside the internal disk plane, its optical interaction projects onto an absolute two-dimensional sky-plane aperture (geometric factor 1.0, independent of the internal aspect ratio η of the lens). The radial gravitational gradient of the baryonic lens attenuates as $a(r) = \frac{GM_{\text{baryon}}}{r^2}$ until it intersects the fundamental cosmological horizon floor a_0:

$$a(R_{BB}) = \frac{GM_{\text{baryon}}}{R_{BB}^2} = a_0 \implies R_{BB} = \sqrt{\frac{GM_{\text{baryon}}}{a_0}} = \sqrt{\frac{2\pi GM_{\text{baryon}}}{cH_0}} \tag{S46}$$

In the perturbed metric $ds^2 = a^2[-(1+2\Psi)d\tau^2 + (1-2\Phi)d\vec{x}^2]$, the null geodesic equation $k^\mu \nabla_\mu k^\nu = 0$ governing the transverse deflection of a light ray with impact parameter $\vec{\xi}$ in the lens plane is:

$$\frac{d^2\vec{\xi}}{dl^2} = -\frac{1}{c^2}\nabla_\perp(\Psi+\Phi) = -\nabla_\perp \ln n_{\text{eff}}(\vec{\xi}, l) \tag{S47}$$

where $n_{\text{eff}} = 1 - \frac{\Psi+\Phi}{c^2}$ is the effective optical refractive index of spacetime. While classical General Relativity integrates continuous weak-field potentials over three-dimensional distances (D_L, D_S, D_{LS}), the DBI saturation factor $\gamma = (1 - 2X/M^4)^{-1/2}$ undergoes a sharp phase transition along the two-dimensional boundary $r_\perp = |\vec{\xi}| = R_{BB}$. This creates a step-like refractive impedance boundary (a 2D topological caustic waveguide) in the lens plane that deterministically routes converging null geodesics around the perimeter R_{BB}, locking the physical Einstein ring radius directly to the saturation boundary:

$$R_E \equiv D_L \theta_E = R_{BB} = \sqrt{\frac{GM_{\text{baryon}}}{a_0}} \tag{S48}$$

where D_L is the angular diameter distance to the lens and θ_E is the observed angular Einstein radius in radians [17–19, 24]. Because in massive elliptical galaxies R_{BB} lies close to the baryonic effective radius ($R_{\text{eff}} \approx R_E$), this phase transition naturally accounts for the apparent departure from a pure 3D power-law profile ($\lambda_{\text{int}} \neq 1$) observed in joint JWST/Keck lensing-kinematic analyses [24], while beyond R_{BB} the continuous radial attenuation $\nu(r) = [1 - \exp(-\sqrt{a_{\text{baryonic}}(r)/a_0})]^{-1}$ governs extended cluster weak-lensing shear profiles out to megaparsec scales [27].

S4.2 Analytical Proof of the Einstein Cross and Fragmented Optical Arcs via Fermat's Principle

By Fermat's principle in General Relativity, observed lensed images form at angular coordinates where the optical arrival-time functional $T(\vec{\theta})$ is stationary ($\nabla_{\vec{\theta}} T(\vec{\theta}) = 0$). Along the saturated boundary perimeter $r_\perp = R_{BB}$, the azimuthal variation of the arrival-time functional for a source aligned with the optical axis is governed by the multipole expansion of the projected baryonic potential $\Phi_{\text{eff}}(R_{BB}, \theta)$ over the azimuthal angle $\theta \in [0, 2\pi)$:

$$\Phi_{\text{eff}}(R_{BB}, \theta) = \Phi_0(R_{BB}) \left[1 + \varepsilon_2 \cos(2\theta) + \sum_{m=3}^{\infty} \varepsilon_m \cos(m(\theta - \theta_m)) \right] \tag{S49}$$

where ε_2 represents the dimensionless baryonic quadrupole ellipticity and ε_m ($m \geq 3$) represent higher-order morphological asymmetries. Imposing Fermat's stationarity condition along the saturation perimeter ($\frac{\partial T}{\partial \theta} \propto -\frac{\partial \Phi_{\text{eff}}}{\partial \theta} = 0$) yields:

$$\frac{\partial \Phi_{\text{eff}}}{\partial \theta} = -\Phi_0(R_{BB}) \left[2\varepsilon_2 \sin(2\theta) + \sum_{m=3}^{\infty} m\,\varepsilon_m \sin(m(\theta - \theta_m)) \right] = 0 \tag{S50}$$

This single equation analytically generates all three observed strong-lensing topologies without dark matter:

1. **Axisymmetric Baryonic Lens ($\varepsilon_2 = 0, \varepsilon_m = 0$):** The azimuthal derivative vanishes identically ($\frac{\partial \Phi_{\text{eff}}}{\partial \theta} \equiv 0$) for all $\theta \in [0, 2\pi)$, focusing light into a complete, continuous Einstein Ring of radius $R_E = R_{BB}$.

2. **Symmetric Elliptical / Quadrupolar Lens ($\varepsilon_2 \neq 0, \varepsilon_{m\geq 3} = 0$):** The stationarity condition reduces to $-2\varepsilon_2\Phi_0 \sin(2\theta) = 0 \implies 2\theta = k\pi$ ($k = 0,1,2,3$), yielding four discrete stationary optical nodes separated by $\pi/2$ (90°): $\theta_1 = 0, \theta_2 = \frac{\pi}{2}, \theta_3 = \pi, \theta_4 = \frac{3\pi}{2}$. Evaluating the second derivative $\frac{\partial^2\Phi_{\text{eff}}}{\partial\theta^2} = -4\varepsilon_2\Phi_0\cos(2\theta)$ confirms that $\theta = 0, \pi$ are arrival-time minima and $\theta = \pi/2, 3\pi/2$ are saddle points, analytically reproducing the quadruple image geometry of the Einstein Cross [19].

3. **Asymmetric / Unrelaxed Baryonic Lens ($\varepsilon_{m\geq 3} \neq 0$):** Localized higher-order multipoles in the visible baryonic mass distribution—including compact, dust-obscured giant molecular clouds (GMCs, $\sim 10^5$–$10^7 M_\odot$ within 10–140 pc), star-forming clumps, and wandering black holes [25, 26]—fragment the stationary locus into discrete angular intervals satisfying $\frac{\partial\Phi_{\text{eff}}}{\partial\theta} \approx 0$, deterministically breaking the R_{BB} perimeter into fragmented optical arcs and localized millilensing perturbations without invoking invisible NFW dark matter subhalos.

Section S5: Statistical Audit Formulation and Primordial Cosmological Continuity (ΛTSM)

S5.1 SI Unit Conversion, Dimensional Homogeneity, and Zero-Parameter 1:1 Goodness-of-Fit

1. Exact SI Kinematic Conversion (SPARC Audit): For a total effective baryonic velocity $V_{\text{baryonic}}(r) = \sqrt{V_{\text{gas}}^2 + \Upsilon_* V_{\text{disk}}^2 + \Upsilon_* V_{\text{bulge}}^2}$ given in km s^{-1} (1 km s^{-1} = 10^3 m s^{-1}) at galactocentric radius r given in kpc (1 kpc = $3.08567758 \times 10^{19}$ m), the exact Newtonian baryonic acceleration in SI units (m s^{-2}) is [20]:

$$a_{\text{baryonic}}(r) = \frac{(10^3 V_{\text{baryonic}}(r))^2}{3.08567758 \times 10^{19}\, r} \tag{S51}$$

2. Strict 1:1 Parity Coefficient of Determination ($R^2_{1:1}$) and p-value Formulation: Unlike standard linear regression, which optimizes two free parameters (slope β_1 and intercept β_0), our zero-parameter audit evaluates model predictions $y_{\text{calc},i}$ directly against the strict identity line $y_{\text{obs},i} = y_{\text{calc},i}$ ($y = x$):

$$R^2_{1:1} \equiv 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^N (y_{\text{obs},i} - y_{\text{calc},i})^2}{\sum_{i=1}^N (y_{\text{obs},i} - \bar{y}_{\text{obs}})^2} \tag{S52}$$

where $\bar{y}_{\text{obs}} = \frac{1}{N}\sum_{i=1}^N y_{\text{obs},i}$. The two-tailed statistical significance (p-value) for sample size N with $\nu_{\text{df}} = N - 2$ degrees of freedom is computed from the exact Student's t-distribution integral:

$$\begin{aligned} t_{\text{stat}} &= r\sqrt{\frac{N-2}{1-r^2}}, \\ p &= 2\int_{|t_{\text{stat}}|}^{\infty} \frac{\Gamma\left(\frac{\nu_{\text{df}}+1}{2}\right)}{\sqrt{\nu_{\text{df}}\pi}\,\Gamma\left(\frac{\nu_{\text{df}}}{2}\right)} \left(1 + \frac{x^2}{\nu_{\text{df}}}\right)^{-\frac{\nu_{\text{df}}+1}{2}} dx \end{aligned} \tag{S53}$$

S5.2 Early-Universe Limit: Covariant Unification with Primordial Cosmology (ΛTSM) and the CMB Acoustic Power Spectra

To demonstrate rigorous theoretical continuity between late-universe galactic equilibrium ($z \approx 0$) and early-universe primordial cosmology ($z \approx 1100$), we evaluate the four-dimensional DBI action and its second-order perturbed Lagrangian (Section S3.1) in a spatially homogeneous FLRW background $ds^2 = a^2(\tau)[-(1+2\Psi)d\tau^2 + (1-2\Phi)\delta_{ij}dx^i dx^j]$ across comoving Fourier wavenumber k [21–23].

1. Background Topological Density Ratio ($\omega_{\text{phantom}}/\omega_b$): In the homogeneous cosmological background ($2\bar{X} \to M^4, \bar{\gamma} \to \infty$), the effective equation of state vanishes ($w_{\text{eff}} = P_\phi/\rho_\phi = \bar{\gamma}^{-1} \to 0$), so the background topological energy density scales identically to pressureless matter ($\rho_{\text{phantom}} \propto a^{-3}$). Equipartition of the 16 directional flux components of the rank-2 metric deformation tensor in the $D = 4$ Lorentzian bulk relative to the $d = D - 1 = 3$ spatial hypersurface degrees of freedom ($\frac{16}{3}$), combined with the one-loop four-dimensional hyperspherical horizon impedance correction ($(4\pi)^{-D/2} = \frac{1}{16\pi^2}$), fixes the exact parameter-free background density ratio:

$$\frac{\omega_{\text{phantom}}}{\omega_b} = \frac{16}{3}\left(1 + \frac{1}{16\pi^2}\right) \approx 5.367134 \tag{S54}$$

For the Planck 2018 baryonic density $\omega_b = 0.02237$ [21], this yields $\omega_{\text{phantom}} = 0.120062$, matching the empirical ΛCDM dark matter density ($\omega_{\text{cdm}} = 0.1200 \pm 0.0012$) to within 0.05% with zero free parameters.

2. Decoupling Acoustic Phase Transition (`tca_off`): During the deep radiation-dominated tight-coupling epoch ($a \ll a_{\text{eq}}$, `tca_on`), photons and baryons are locked into a single relativistic fluid ($R \gg 1, \theta_b \approx \theta_\gamma$), preventing independent baryonic slip. As the plasma enters the decoupling transition (`tca_off`, $a \sim a_{\text{eq}} \to \pi a_{\text{eq}}$), longitudinal baryonic compression modes couple to the trace-free 3D Maclaurin spatial shear operator ($\mathcal{K}_0 = 1 - \frac{1}{3} = \frac{2}{3}$) and the causal horizon surface impedance ($\Gamma_0 = \frac{1}{8\pi}$). Enforcing strict super-horizon causality via the Helmholtz–Yukawa projection operator $\Pi_{\text{causal}}(k,\mathcal{H}) = \frac{k^2}{k^2+\mathcal{H}^2}$ and post-recombination neutralization freeze-out via $\Theta_{\text{rec}}(a) = \exp\left[-\left(\frac{a}{\pi a_{\text{eq}}}\right)^4\right]$ yields the parameter-free baryonic Euler momentum equation:

$$\begin{aligned} &\theta_b' + [\mathcal{H} + \Gamma_{\text{active}}(k,a,\theta_b)]\,\theta_b - \left[c_s^2 k^2 - \mathcal{K}_{\text{phantom}}(k,a)\right]\delta_b \\ &\quad - k^2\Psi - R\tau_c^{-1}(\theta_\gamma - \theta_b) = 0 \end{aligned} \tag{S55}$$

where:

$$\mathcal{K}_{\text{phantom}}(k,a) = \frac{2}{3}\mathcal{H}^2\left(\frac{k^2}{k^2+\mathcal{H}^2}\right)\exp\left[-\left(\frac{a}{\pi a_{\text{eq}}}\right)^4\right] \tag{S56}$$

$$\Gamma_{\text{active}}(k,a,\theta_b) = \frac{1}{8\pi}\left(\frac{a_{\text{eq}}}{a}\right)\left(\frac{k^2}{k^2+\mathcal{H}^2}\right)\exp\left[-(a/\pi a_{\text{eq}})^4\right]\Theta(\theta_b) \tag{S57}$$

- **In the Decoupling Primordial Regime ($a_{\text{eq}} \lesssim a \lesssim \pi a_{\text{eq}}, z \approx 3400 \to 1100$):** The elastic curvature term $\mathcal{K}_{\text{phantom}}$ stabilizes the acoustic gravitational potential wells, while the directional horizon braking Γ_{active} regulates rarefaction peaks, reproducing the Planck 2018 CMB TT, TE, EE power spectra ($R^2_{TT} = 0.9970, R^2_{TE} = 0.9951, R^2_{EE} = 0.9951$) without cold dark matter.

- **In the Post-Recombination and Late-Universe Regime ($a \gg \pi a_{\text{eq}}, z \to 0$):** The plasma phase-transition

gate $\Theta_{\rm rec}(a)$ freezes out acoustic damping, preserving the exact linear matter clustering amplitude ($\sigma_8 = 0.8229$) on large scales while reducing continuously to the quasi-static spatial saturation equations (Sections S1–S4) within virialized galactic and cluster structures below $a_0 = \frac{cH_0}{2\pi}$.

References

[1] G. W. Gibbons and S. W. Hawking, "Cosmological Event Horizons, Thermodynamics, and Particle Creation," *Phys. Rev. D* **15**(10), 2738–2751 (1977).

[2] T. Jacobson, "Thermodynamics of Spacetime: The Einstein Equation of State," *Phys. Rev. Lett.* **75**(7), 1260–1263 (1995).

[3] A. G. Riess *et al.*, "Large Magellanic Cloud Cepheid Standards Provide a 1% Foundation for the Determination of the Hubble Constant," *Astrophys. J.* **876**(1), 85 (2019).

[4] A. G. Riess *et al.*, "A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s^{-1} Mpc^{-1} Uncertainty from the Hubble Space Telescope and the SH0ES Team," *Astrophys. J. Lett.* **934**(1), L7 (2022).

[5] R. P. van der Marel *et al.*, "The M31 Velocity Vector. II. Radial Orbit Toward the Milky Way and Implied Local Group Mass," *Astrophys. J.* **753**(1), 8 (2012).

[6] P. van Dokkum *et al.*, "A galaxy lacking dark matter," *Nature* **555**(7698), 629–632 (2018).

[7] M. Born and L. Infeld, "Foundations of the New Field Theory," *Proc. R. Soc. Lond. A* **144**(852), 425–451 (1934).

[8] P. A. M. Dirac, "An Extensible Model of the Electron," *Proc. R. Soc. Lond. A* **268**(1332), 57–67 (1962).

[9] M. Alishahiha, E. Silverstein, and D. Tong, "DBI in the Sky: Non-Gaussianity from Inflation with a Speed Limit," *Phys. Rev. D* **70**(12), 123505 (2004).

[10] B. P. Abbott *et al.* [LIGO Scientific and Virgo Collaborations], "Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848**(2), L13 (2017).

[11] C.-P. Ma and E. Bertschinger, "Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges," *Astrophys. J.* **455**, 7–25 (1995).

[12] J. Garriga and V. F. Mukhanov, "Perturbations in k-Inflation," *Phys. Lett. B* **458**(2–3), 219–225 (1999).

[13] C. C. Lin and F. H. Shu, "On the Spiral Structure of Disk Galaxies," *Astrophys. J.* **140**, 646–655 (1964).

[14] S. S. McGaugh, J. M. Schombert, G. D. Bothun, and W. J. G. de Blok, "The Baryonic Tully–Fisher Relation," *Astrophys. J. Lett.* **533**(2), L99–L102 (2000).

[15] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Phys. Rev. Lett.* **117**(20), 201101 (2016).

[16] C. M. Will, "The Confrontation between General Relativity and Experiment," *Living Rev. Relativ.* **17**(1), 4 (2014).

[17] A. S. Bolton *et al.*, "The Sloan Lens ACS Survey. I. A Large Spectroscopically Selected Sample of Massive Early-Type Lens Galaxies," *Astrophys. J.* **638**(2), 703–724 (2006).

[18] A. S. Bolton *et al.*, "The Sloan Lens ACS Survey. V. The Full ACS Strong-Lens Sample," *Astrophys. J.* **682**(2), 964–989 (2008).

[19] T. Treu, "Strong Lensing by Galaxies," *Annu. Rev. Astron. Astrophys.* **48**, 87–125 (2010).

[20] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astron. J.* **152**(6), 157 (2016).

[21] Planck Collaboration, N. Aghanim *et al.*, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020).

[22] J. Lesgourgues, "The Cosmic Linear Anisotropy Solving System (CLASS) I: Overview," `arXiv:1104.2932` (2011).

[23] D. Blas, J. Lesgourgues, and T. Tram, "The Cosmic Linear Anisotropy Solving System (CLASS) II: Approximation schemes," *J. Cosmol. Astropart. Phys.* **2011**(07), 034 (2011).

[24] T. Treu, "Cosmology with strong gravitational lenses," `arXiv:2610.19598` (2026).

[25] H. R. Stacey, J. P. McKean, M. Pandey-Pommier, C. Cornil-Baïotto, S. Vegetti *et al.*, "Tracing small-scale structure formation through cosmic time with strong gravitational lensing," `arXiv:2610.20710` (2026).

[26] A. Zanella, A. L. Faisst, R. Herrera-Camus, E. Chien *et al.*, "ALMA2040: Understanding the Early Assembly of Galaxies — The Formation of Star-Forming Clumps and their Role in Shaping Galaxy Evolution," `arXiv:2610.20730` (2026).

[27] J. M. Diego, G. Congedo, R. Gavazzi, T. Schrabback *et al.* [Euclid Collaboration], "Euclid: Early Release Observations — A combined strong and weak lensing solution for Abell 2390 beyond its virial radius," *Astron. Astrophys.* **706**, A83 (2026).